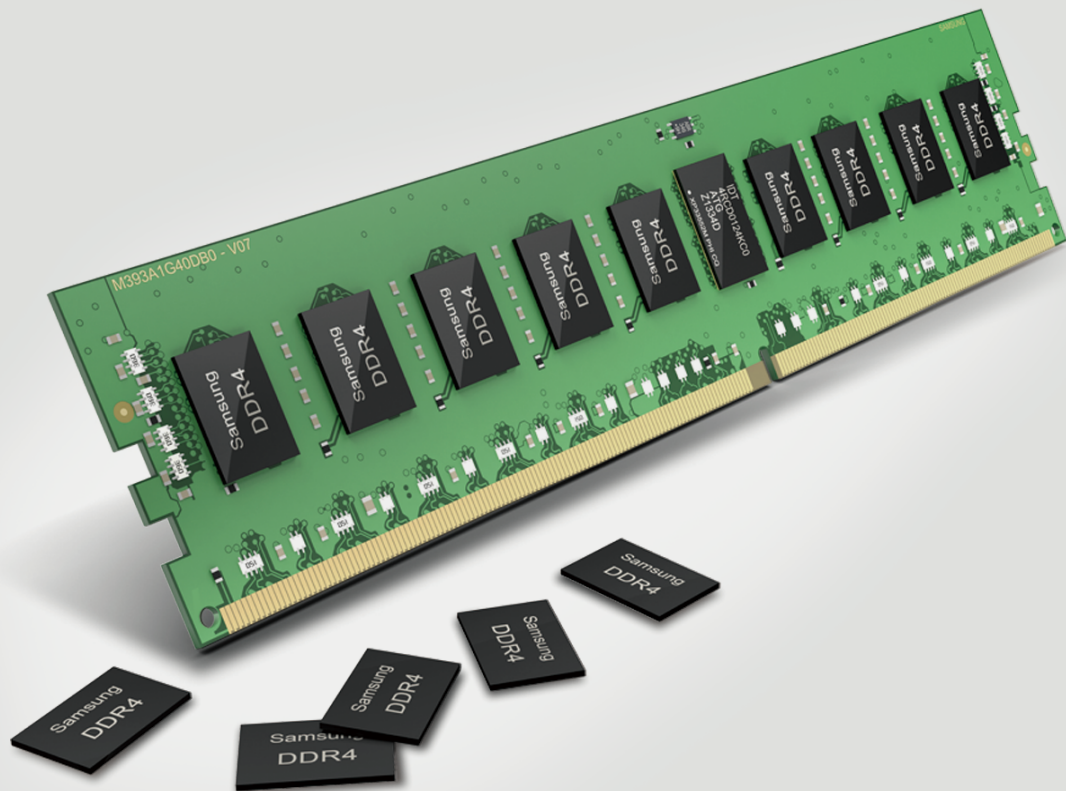


Samsung Memory DDR4 SDRAM

The new generation of high-performance, power-efficient memory that delivers great reliability for enterprise applications



An optimized memory for enterprise-level workloads and client

Meet diverse enterprise workload demands, while enhancing client performance and battery life with higher bandwidth and reduced power consumption

Accelerated adoption of cloud computing, virtualization and high-performance computing (HPC) technologies has made higher-performing, higher-density memory a key factor for optimal server operations. Highly virtualized environments enable companies to run numerous applications on a single server instead of multiple servers. A single server with more virtual machines (VMs) requires not only a higher processor speed, but also higher memory density .

Requirements for memory become more diverse when supporting a wide range of enterprise server applications, from less critical workloads to mission-critical workloads. Enterprise-level workloads, such as databases or transaction processing that run on high-end servers, need a large capacity of in-memory systems and higher reliability.

Mid-range servers that are used for virtualization or consolidation require high bandwidth and scalability. Small form factor, lower power consumption and low cost are essential requirements for

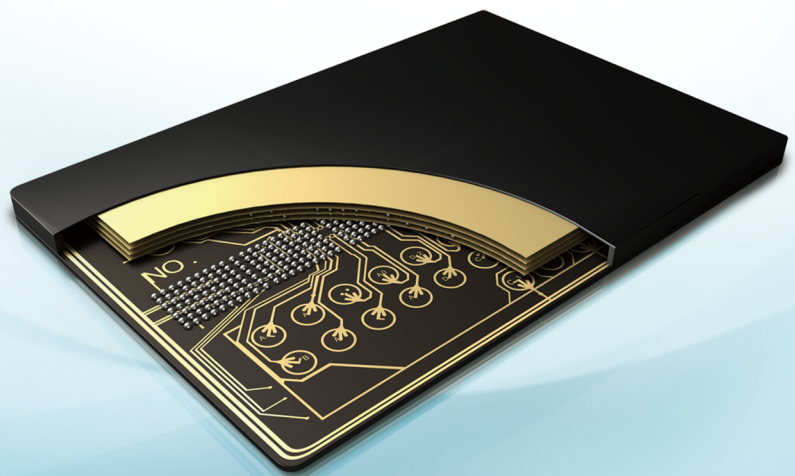
workloads on low-end servers used as web, collaboration and infrastructure systems.

Rising energy costs and the need to provide greater environmental sustainability are also placing demands on chipmakers and server vendors. A server running a virtualized environment can achieve a higher utilization that, in turn, increases the total power consumption of the server. As a result, CPU and server companies are focusing more on the development of next-generation green IT systems.

The memory that supports next-generation, green IT systems should meet the diverse demands of enterprise workloads with higher performance, increased density, improved reliability and low power consumption.

As screen resolution increases (up to 8K UHD), more memory bandwidth is also needed for client applications. Adopting higher bandwidth memory, users can better experience the more natural and smoother texture inherent in high resolution screens. Also, one of the most important factors in portable PCs is battery life time. By reducing memory power consumption, users can take advantage of their portable PCs longer with extended battery life.

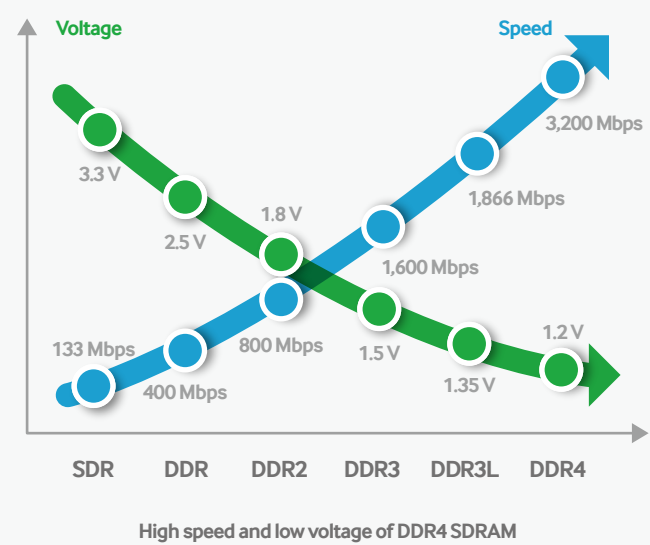
Samsung Memory 3DS TSV DRAM Package



Leverage the expertise of a technology leader in innovation

Samsung DDR4 SDRAM provide an optimized solution for enterprise applications

Samsung DDR4 is an optimized solution for highly virtualized environments, high-performance computing and networking as well as PC clients. Samsung DDR4-based modules are designed with a new system circuit architecture to deliver higher performance with lower power requirements than previously available memory products.



The Samsung portfolio of DDR4-based modules using 20 nm-class process technology includes registered dual inline memory modules (RDIMMs), load-reduced DIMMs (LRDIMMs) for servers and unregistered DIMMs (UDIMMs), and small outline DIMMs (SODIMMs) for workstations and PCs. These memory modules are available with initial speeds up to 2667 Mbps, increasing to the Joint Electron Devices Engineering Council (JEDEC)-defined 3200 Mbps.

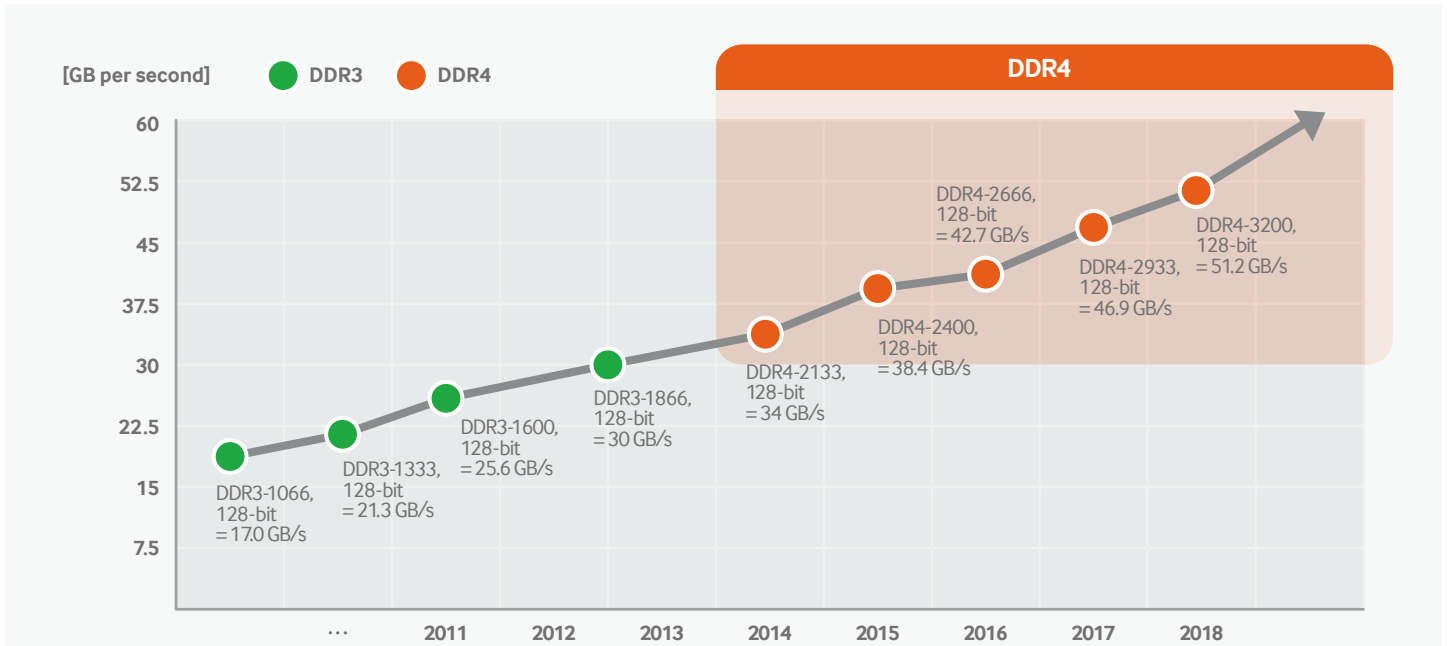
The portfolio includes the following modules :

Density	Enterprise	Client
4 GB	-	UDIMMs/SoDIMMs
8 GB	RDIMMs	UDIMMs/SoDIMMs
16 GB	RDIMMs	UDIMMs/SoDIMMs
32 GB	RDIMMs/LRDIMMs	UDIMMs/SoDIMMs
64 GB	RDIMMs/LRDIMMs	-
128 GB	RDIMMs/LRDIMMs	-
256 GB	RDIMMs	-

Doubled bandwidth, along with reduced voltage and dramatically lower power consumption, improve performance and optimize the total cost of ownership (TCO). Samsung DDR4's enhanced reliability, availability and serviceability (RAS) features provide enhanced reliability and improved signal integrity (S/I).

To meet the ever increasing demand for higher density server memory modules, Samsung offers 3DS (3 Dimensional Stacked) DDR4 RDIMMs which apply TSV (Through Silicon Via) technology. The new TSV 3DS stack technology enables higher density DRAM stack packages with higher operational frequency and reduced power consumption. The RDIMM configuration along with 3DS TSV DRAM packages exhibits performance benefits and lower power consumption over alternative high density solutions to ultimately provide the lowest TCO for data centers.

Low - power memory consumption with higher capacity and performance

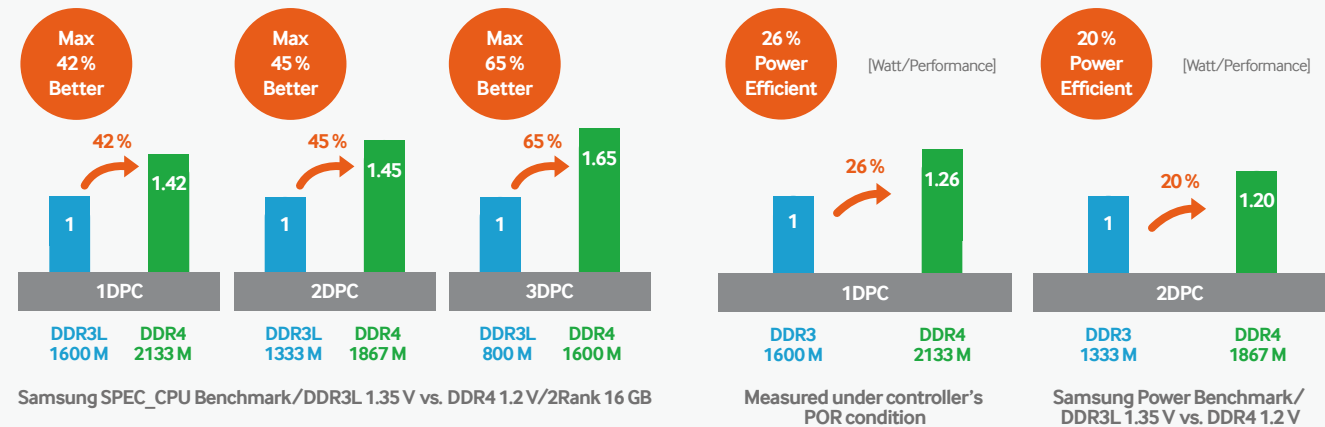


Increased performance for higher bandwidth

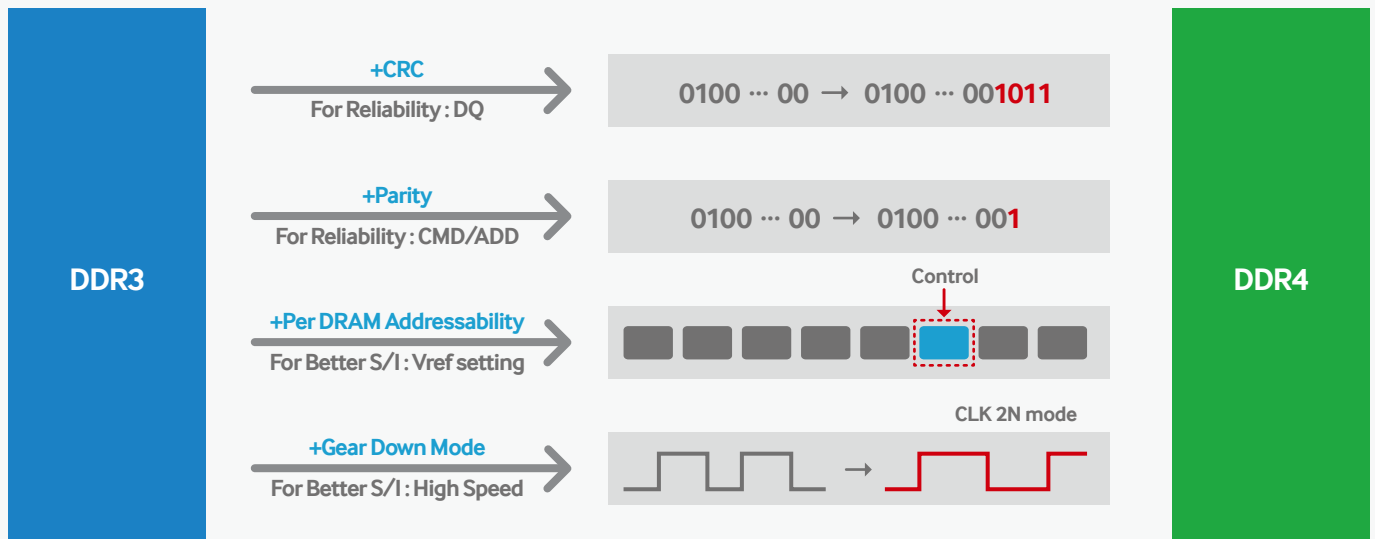
Samsung DDR4 delivers higher performance at higher speeds than DDR3. DDR4 can achieve up to 3.2 Gbps bandwidth. Compared to DDR3L (low power DDR3), Samsung DDR4 shows overall performance improvements in every DIMM and over 40% better performance at 1DIMM per channel.

Reduce power usage for greener, lower-cost computing

A major decrease in voltage and the improved input/output (I/O) power efficiency of DDR4 synchronous dynamic RAM (SDRAM) translates into significant cost savings. According to Samsung internal testing, DDR4 boasts a continuous working voltage of 1.2V, a voltage that is 11% lower than the 1.35V consumed by DDR3L. It also has a Pseudo Open Drain (POD) interface to reduce I/O power consumption and increase power savings by 50% compared to DDR3L. Overall DDR4 shows a higher performance/watt of over 20%



100% detection of random 1- to 2-bit errors



RAS feature comparison of DDR3 and DDR4 SDRAM

Provide greater reliability, availability and serviceability (RAS)

DDR3 modules provided only one RAS feature, their error-correcting code (ECC) capability. Compared to DDR3, Samsung DDR4 provides more robust RAS features, such as Cyclic Redundancy Check (CRC), Parity, Per DRAM Addressability and Gear Down Mode.

Enhanced reliability and improved S/I for mission-critical enterprise applications are directly attributable to Samsung DDR4's significant advancements with the following RAS features:

1. Cyclic Redundancy Check (CRC) for improved data reliability

CRC is error-detecting code that detects accidental changes to raw data of DRAM's DQ. CRC confirms 100% detection of random 1- to 2-bit errors by enabling error detection capability for data transfer.

2. On-chip parity detection for the integrity of Command/Address

Parity for Command/Address (CMD/ADD) provides a method of verifying the integrity of command and address transfers over a link.

3. Per DRAM Addressability for enhanced signal integrity

Samsung DDR4 can control module components and enhance signal integrity by controlling the ODT of Vref levels.

4. Gear Down Mode for improved signal integrity

DDR4 Gear Down Mode allows a high speed of DQ to be maintained while decreasing the high speed of CMD/ADD.

Fault-tolerant, higher RAS systems supported by DDR4 SDRAM can remain available for considerably longer periods of time without failure.

Samsung DDR4 SDRAM

Manage a range of enterprise workloads with greater reliability, doubled bandwidth and reduced power consumption

Designed with advanced system circuit architecture, Samsung DDR4 supports a wide range of server memory needs by

delivering higher performance and reduced power consumption with increased reliability. By taking advantage of the advanced features of Samsung DDR4, companies can achieve greater performance at a lower TCO.

DDR3 and DDR4 Specifications and features comparison

Feature		DDR3	DDR4
Component density, speed		512 Mb - 4 Gb	4 Gb - 16 Gb
		0.8 - 1.6 Gbps	1.6 - 3.2 Gbps
Module density		1, 2, 4, 8, 16, 32 and 64 GB	8, 16, 32, 64 and 256 GB
Interface	Voltage(VDD, VDDQ, VPP)	1.5V, 1.5V, NA (1.35V, 1.35V, NA)	1.2 V, 1.2 V, 2.5 V
	Vref	External Vref (VDD, 2)	Internal Vref (need training)
	Data I/O	Center Tab Termination (CTT) (34 ohm)	POD (34 ohm)
	CMD, ADDR I/O	CTT	CTT
	Strobe	Bi-dir, diff	Bi-dir, diff
Core architecture	Number of banks	8 banks	16 banks (4-bank group)
	Page size (X4, 8, 16)	1 KB, 1 KB, 2 KB	512 B, 1 KB, 2 KB
	Number of prefetch	8 bits	8 bits
	Added functions	RESET, ZQ, Dynamic ODT	RESET, ZQ, Dynamic ODT, CRC, Data Bus Inversion (DBI), Multiple preamble
Physical	Package type, balls (X4, 8, X16)	78, 96 BGA	78, 96 BGA
	DIMM type	R, LR, U, SoDIMM	R, 3DS R, LR, (ECC/nECC) U/SoDIMM
	DIMM pins	240 (R, LR, U), 204 (So)	288 (R, LR, U), 260 (So)

Legal and additional information

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For more information

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